https://doi.org/10.1038/s44335-025-00047-z

AI-native robotic vision systems enabled by in-sensor computing

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Artificial intelligence (AI) is accelerating the evolution of robotics from task-specific automation to general-purpose autonomy, enabling robots to perform high-level tasks in unstructured and dynamic environments. One of the key enablers in this evolution is the integration of AI with robotic vision systems, which provide accurate perception and contextual interpretation of complex surroundings. An important challenge for this goal is to ensure computational efficiency while robust inference is achieved. One potential solution to tackle this challenge is the acquisition of visual data in formats inherently optimized for AI computing—a concept referred to as AI-native robotic vision. In this review, we highlight recent developments in robotic vision systems with in-sensor computing capabilities. We first discuss the functional features of synapses, neurons, and retinal hierarchies in biological vision systems, and introduce in-sensor computing techniques enabled by device-level emulation of such features. We then present representative studies for each class of in-sensor computing techniques, highlighting their operation principles and robotic vision applications. Finally, we discuss future research directions for advancing in-sensor computing techniques toward the practical implementation of AI-native robotic vision.

Artificial intelligence (AI) has fundamentally transformed numerous industries, and is now driving a significant paradigm shift in robotics—from traditional robots operating under predefined instructions in structured environments to AI-integrated robots capable of human-like autonomy in unstructured and dynamic conditions[1]. Traditional robots were typically limited to repetitive, low-level tasks that are simple for humans but undesirable to perform, such as component assembly in manufacturing lines (Fig. 1a, (i)) or basic household assistance (Fig. 1a, (ii)). In high-precision scenarios (e.g., surgery), they merely served as assistive tools fully controlled by human operators (Fig. 1a, (iii)).

Today, with the integration of AI, robots are capable of performing general, high-level tasks in complex environments beyond hard-coded instructions[2,3]. By interpreting situational context and generating adaptive action plans through AI-driven decision making, robots can operate with human-like intelligence and adaptability[4,5]. In industrial fields, they recognize dynamic situations and proactively execute diverse tasks (Fig. 1b, (i))[6]. At home, they understand user intent and autonomously carry out a wide range of context-aware activities (Fig. 1b, (ii))[7]. In healthcare, they can assess patient status, make diagnostic decisions, and even perform surgical procedures with minimal human intervention (Fig. 1b, (iii))[8–11].

A key enabler of such AI-integrated robotic capabilities is vision systems[12–16]—human also acquires most of the information through vision. Among various sensing modalities, robotic vision is particularly effective for acquiring task-relevant information[17,18]. Furthermore, recent advances in visual perception AI models—especially in object detection and contextual interpretation—have enabled the extraction of semantically meaningful features from visual data[19–21]. However, these capabilities have come at the cost of increased model complexity and computational demands, which cannot meet the requirements of mobile robotic platforms constrained by power, memory, and latency[22–25]. As a result, recent efforts have shifted toward optimizing system-level efficiency while maintaining high levels of AI performance.

To meet the efficiency demands, it is not sufficient to optimize AI model architectures alone. It is equally important to optimize how visual data is acquired and formatted to align with the requirements of AI computing[26,27]. This approach defines the concept of 'AI-native robotic vision', which aims to generate visual data in formats inherently optimized

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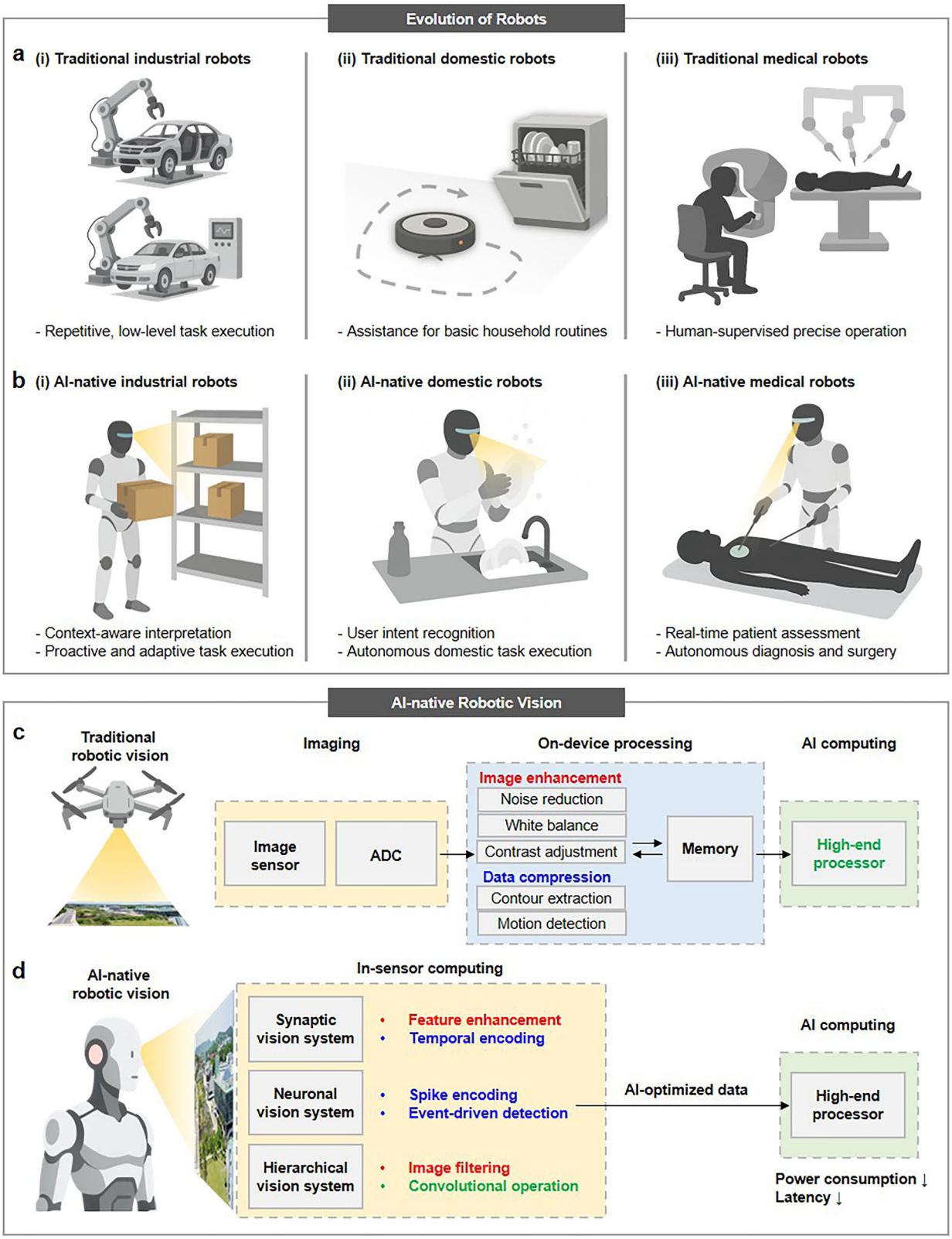


Fig. 1 | AI-native robotic vision. a Schematic illustration of traditional robots operating under predefined instructions in structured environments: industrial robots performing repetitive, low-level tasks (i), domestic robots assisting with simple household routines (ii), and medical robots serving as human-supervised surgical tools (iii). **b** Schematic illustration of AI-native robots equipped with human-like autonomy for unstructured environments: industrial robots capable of context-aware interpretation and adaptive task execution (i), domestic robots recognizing user intent and autonomously performing household tasks (ii), and medical robots capable of real-time patient assessment, diagnosis, and surgery with minimal human intervention (iii). **c** Imaging and data processing pipelines of traditional robotic vision systems, where raw images captured by image sensors undergo multiple preprocessing stages before being transferred to external high-end processors for AI inference. **d** Streamlined imaging and data-processing pipeline of AI-native robotic vision systems, where synaptic, neuronal, and hierarchical vision systems generate AI-optimized data suitable for AI inference.

for AI inference, minimizing the need for intermediate processing without compromising AI performance. However, conventional robotic vision systems, which employ modular architecture with physically separated image sensors and processors, are not ideal for this goal[28–32]. In such systems, raw visual data should undergo multiple image signal processing stages—including noise reduction, white balance, contrast adjustment, contour extraction, and motion detection—to be converted into AI-optimized formats (Fig. 1c)[33,34]. These additional preprocessing steps introduce processing latency and increase power consumption[35–37].

In-sensor computing offers a promising solution by enabling key computational operations, such as feature enhancement, spike encoding, and convolutional filtering, to be performed directly at the sensory level[38]. By executing these operations at the point of data acquisition, the image signal processing pipeline is streamlined and redundant data transmission is eliminated, thereby reducing both power consumption and latency (Fig. 1d)[39,40]. Moreover, in-sensor computing facilitates the direct generation of AI-optimized visual data, including feature-enhanced images, spike-encoded signals, and compressed feature maps, without the need for additional on-device processing. This not only enhances system-level efficiency but also improves the robustness of AI inference in robotic applications[41–43].

The operating principles of in-sensor computing are inspired by neural signal processing mechanisms of biological vision systems. In the retina, visual information undergoes substantial neural processing through hierarchical networks of synapses and neurons, significantly reducing the computational burden on downstream brain processing[44,45]. Here, we summarize recent advances in in-sensor computing techniques for enabling AI-native robotic vision with a focus on three biologically inspired motifs: synaptic, neuronal, and hierarchical characteristics. Synaptic vision systems, emulating spatiotemporal integration behavior of synapses, selectively amplify meaningful features while filtering out noise—suitable for feature enhancement and temporal encoding[46]. Neuronal vision systems replicate the spike-based signaling of neurons, converting analog stimuli into discrete spike trains compatible with spiking neural networks (SNNs) or event-driven AI models[47]. Hierarchical vision systems, mimicking the layered architecture of the retina, enable spatial feature extraction and convolutional operation, reducing data volume and thus alleviating transmission bottlenecks[48]. Finally, we outline the key device-level and system-level requirements for translating these in-sensor computing techniques from the laboratory.

Synaptic vision systems for AI-native robotic vision

In biological vision systems, synapses function not only as junctions for neural signal transmission but also as dynamic computational elements. At synapses, incoming neural signals from presynaptic neurons are integrated over time, and the strength of signal integration (i.e., synaptic weight) is dynamically modulated by synaptic plasticity (e.g., short-term plasticity (STP), long-term plasticity (LTP), and spike-timing-dependent plasticity). With such attributes, neural signals are simultaneously processed during their propagation pathway through synapses.

Synaptic vision systems emulate those signal-integrative attributes to implement synapse-like computing functionalities at the sensory level[49]. In this section, we discuss two key computing functionalities of synaptic vision systems, including feature enhancement that selectively amplifies semantically important visual information and temporal encoding that transforms time-dependent visual streams into compact data suitable for AI inference. We also present representative examples of synaptic vision systems with their applications to AI-native robotic vision.

Feature enhancement by synaptic vision systems

Synapses transmit neural signals from presynaptic to postsynaptic neurons by converting presynaptic action potentials into excitatory postsynaptic potentials (EPSPs)[50–53]. Upon the arrival of action potentials, neurotransmitters (*e.g.*, glutamate) are released into the synaptic cleft and bind to receptors on the postsynaptic membrane, generating EPSPs (Fig. 2a)[39,54]. These EPSPs are temporally integrated through the repeated presynaptic stimulation, and also spatially integrated through simultaneous inputs arriving at multiple synapses converging onto the same postsynaptic neuron[55,56]. This spatiotemporal integration enables synapses to act as the first computational stage of neural processing, selectively transmitting signals that are temporally or spatially reinforced[57,58].

Synaptic vision systems emulate those integrative characteristics of synapses by employing synaptic photodetectors, which generate output signals depending on the total light dose—a product of light intensity and exposure duration—rather than responding instantaneously like conventional photodetectors (e.g., Si photodiode)[59–61]. This dose-dependent photoresponse enables the integration of visual stimuli at the sensory level, where stronger or longer irradiation results in higher output signals. With such attributes, synaptic photodetectors can selectively amplify semantically meaningful signals while filtering out transient or low-relevance noise (Fig. 2b)[60,62]. Consequently, synaptic vision systems perform feature enhancement processing at the sensory level, such as noise reduction, contrast enhancement, and salient attention, producing AI-optimized visual data for accurate downstream interpretation[63–68].

For instance, synaptic vision systems can suppress temporally inconsistent signals, such as random background noise (e.g., salt-and-pepper noise), while selectively extracting light patterns with frequent or sustained exposure (Fig. 2c, (i))[69]. It was achieved through their synapse-inspired integrative characteristics, where the output photocurrent was proportional to the total light dose. In this device, transient inputs generated minimal photocurrent, whereas repeated stimuli resulted in large photocurrent accumulation. Therefore, the signal-to-noise ratio increased, enhancing overall image contrast and thus improving perceptual clarity.

Nevertheless, the abovementioned synaptic photodetectors still suffer from residual noise, which can lead to false recognition results when processed by noise-sensitive deep neural networks (DNNs). To address this limitation, Seung et al. applied threshold-filtering functions to synaptic photodetectors by integrating them with diodes exhibiting threshold-switching characteristics[70]. This device cut off the subthreshold outputs—typically generated by background noise—while retaining only meaningful signals that usually exceed the threshold (Fig. 2c, (ii)). Consequently, the signal-to-noise ratio was substantially enhanced, enabling the acquisition of noise-free images. When evaluated using the ResNet-50 DNN model, these noise-free images achieved a recognition accuracy of 86%, notably higher than the 50% accuracy obtained from the conventional synaptic photodetectors without thresholding functionality.

In real-world environments, illumination conditions vary widely, from dimly lit scenes to brightly illuminated settings, necessitating adaptive contrast enhancement for consistent visual perception. To meet this need, Liao et al. developed a synaptic vision system using a MoS_2 phototransistor, in which the degree of signal integration could be adaptively modulated via gate bias control (Fig. 2c, (iii))[71]. By tuning the gate voltage, the device adjusted its temporal integration behavior to suit ambient lighting—amplifying weak signals under low-light conditions and suppressing oversaturation in bright conditions. This adaptability enabled the extraction of semantically meaningful features with consistently high contrast, thereby improving the recognition accuracy of AI models across diverse lighting conditions.

Recently, synaptic vision systems extended their in-sensor computing functionalities to color contrast enhancement. For instance, a synaptic phototransistor based on Zr-doped $CsPbI_3$ nanocrystals selectively highlighted blue-dominant features in visual scenes through wavelength-selective plasticity transition from volatile STP to nonvolatile LTP (Fig. 2c, (iv))[72]. This device showed LTP responses under blue light illumination (405 nm), while exhibiting STP responses under the irradiation of longer-wavelength light (e.g., 650 nm). As a result, the blue-dominant objects were captured with salient attention, while features in other colors were weakly represented, enabling selective recognition of color-salient targets.

In another example, Deng et al. demonstrated motion blur correction using a synaptic vision system that enhanced the clarity of structural edges

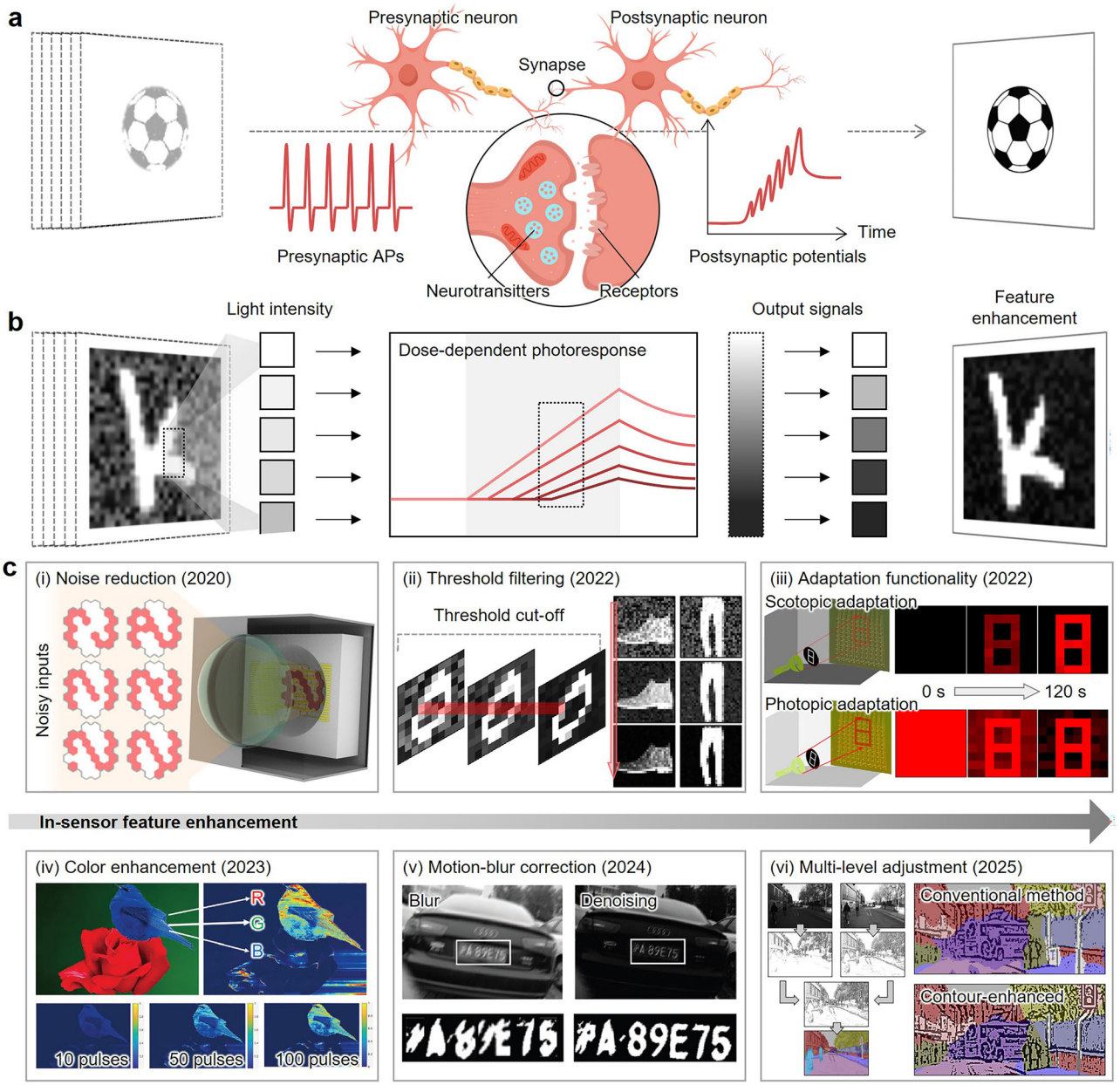


Fig. 2 | Feature enhancement by the synaptic vision system. a Schematic illustration of a biological synapse that transmits neural signals from presynaptic neurons to postsynaptic neurons. EPSPs are integrated upon the arrival of presynaptic action potentials. **b** Schematic illustration showing feature enhancement by synaptic vision systems. The dose-dependent photoresponse of synaptic photodetectors enables noise reduction and contrast enhancement. **c** Representative demonstrations of feature enhancement by synaptic vision systems. The synaptic vision systems reduced background noise and enhanced image contrast using synapse-inspired integrative characteristics (i), with performance further improved by incorporating threshold filtering (ii) and adaptive functionality (iii). Beyond these capabilities, synaptic vision systems also enabled wavelength-selective color enhancement (iv), motion-blur correction via contour-selective accumulation (v), and multi-level image adjustment for high-clarity contour extraction (vi). (i) Reproduced under the terms of the Creative Commons CC-BY 4.0 license[69]. Copyright 2020, Springer Nature. (ii) Reproduced under the terms of the Creative Commons CC-BY 4.0 license[70]. Copyright 2022, American Association for the Advancement of Science. (iii) Reproduced with permission[71]. Copyright 2022, Springer Nature. (iv) Reproduced with permission[72]. Copyright 2023, Wiley-VCH. (v) Reproduced with permission[73]. Copyright 2024, Wiley-VCH. (vi) Reproduced under the terms of the Creative Commons CC-BY 4.0 license[74]. Copyright 2025, American Association for the Advancement of Science.

through contour-selective signal accumulation[73]. In this system, the synaptic photodetector accumulated the photocurrent from sharp spatial features, such as the contours of license plates, that generated temporally consistent optical inputs. In contrast, blurred regions with temporally irregular inputs resulted in much weaker signal accumulation. As a result, the device acquired high-contrast images with continuous, sharply defined contours (Fig. 2c, (v)). Compared to raw blurred inputs that yielded faint and fragmented contours after thresholding, the blur-corrected images preserved salient object boundaries. This, in turn, enhanced visual clarity and improved object recognition performance under motion-blurred conditions.

The feature enhancement capability of synaptic vision systems also enables high-clarity contour extraction, enhancing both the accuracy and efficiency of semantic segmentation in automatic driving scenarios[74]. AI computing can significantly reduce the load for data transmission and computation by relying solely on structural contours rather than full-

resolution raw images. To achieve high segmentation accuracy, however, clearly defined object boundaries are essential, which are difficult to extract from a single image because bright and dark regions typically coexist in real-world environments. Kwon et al., therefore, captured two complementary images optimized for both bright and dim regions by adaptively modulating the integration behavior of a synaptic phototransistor via gate bias control (Fig. 2c, (vi)). By extracting contours from each image and merging them, high-clarity contour extraction was achieved, enabling semantic segmentation accuracy comparable to that of high-resolution inputs while reducing the data volume to 8.8% of the original.

Temporal encoding by synaptic vision systems

The primary role of biological synapses is not simply relaying neural signals but transforming presynaptic inputs into refined outputs that downstream neural networks can effectively process[44]. In synapses, this is achieved by temporal summation of repeated presynaptic stimulation and spatial summation of multi-synapse inputs (Fig. 3a)[75–77]. With such attributes, synapses generate postsynaptic responses that embed the temporal continuity and spatial correlation of incoming stimuli, encoding perceptually meaningful signals that involve informative features while suppressing redundancy[78–81].

Synaptic vision systems emulate temporal encoding capabilities at the sensory level, converting continuous streams of optical inputs into compact data optimized for downstream AI computing[82–85]. This encoding strategy is particularly advantageous for resource-constrained AI models, as it reduces data volume and network parameters while preserving salient information[86–88]. In reservoir computing, synaptic vision systems compress high-dimensional visual streams into low-dimensional yet information-preserving data by generating distinct output states according to the temporal sequence of optical pulses (Fig. 3b)[89–91]. By preserving key spatial features while substantially lowering data volume, both computational load and transmission bottlenecks for AI computing are mitigated[92].

For example, synaptic photodetectors can encode spatial segments of an image into single output values, thereby reducing data dimensionality and thus enhancing the overall computing efficiency[93]. This encoding process begins by converting each spatial segment into a temporally programmed sequence of optical pulses, which is then applied to the synaptic photodetectors. Owing to their memory and decay characteristics, the devices can accumulate photocurrents differently depending on the temporal input patterns, thereby generating distinct output values that reflect the original spatial information (Fig. 3d, (i)). For instance, a 5×5 image that would typically require 25 input nodes can be compressed into just five outputs, each representing one segment, via reservoir computing. This data dimension compression enables AI models to operate with lower computational load and fewer parameters.

To enhance recognition accuracy of the compressed data, Zhang et al. proposed a dual sampling strategy that extracted two complementary output values from each spatial segment[94]. While reservoir computing compresses spatial patterns into single output values, the reduction in data dimensionality can lead to ambiguity between spatial patterns, increasing the risk of misclassification. To overcome this limitation, output values were sampled at two different time points—once after a short delay and once after a longer delay—following the irradiation of temporally encoded optical pulses (e.g., 4-bit on/off pulses) (Fig. 3d, (ii)). These two sampled values provided complementary information for each spatial segment, allowing AI models to distinguish the spatial patterns more reliably and thereby achieve higher recognition performance.

Recently, reservoir computing has extended encoding capabilities from single-modality vision to multimodal perception, enabling the integrated processing of heterogeneous sensory information. For instance, reconfigurable synaptic phototransistors that simultaneously respond to both optical and electrical inputs were developed to handle visual and auditory signals together (Fig. 3d, (iii))[95]. By assigning light pulses to vision and voltage pulses to sound, the device generated cross-modal features that preserved the correlations across sensory domains. Therefore, this multimodal encoding provided information-rich and complementary features, reducing the ambiguities inherent in single-modality computing while compacting data volume.

Beyond static encoding, synaptic vision systems can also encode dynamic motion (Fig. 3c). For example, synaptic photodetectors exhibiting intrinsically slow photocurrent decay can encode motion trajectories within a single image by retaining afterimages of sequential stimuli[96]. When an object moves across the device array, photocurrents at its previous positions remain visible as fading afterimages, generating a spatiotemporal contour of motion in which recent positions appear bright while earlier ones gradually fade (Fig. 3d, (iv)). By analyzing these residual traces, AI models can infer the path and direction of motion from a single frame, eliminating the need for frame-by-frame video analysis.

Recent work utilized synaptic vision systems to achieve semantic data compression[97]. Instead of storing motion as residual traces, this system integrated spatiotemporal information of an entire video into a single binary-coded frame. During this process, strong and transient signals corresponding to salient moving objects were selectively amplified, while weak and redundant background was suppressed, yielding a compact image where motion is directly encoded in high-contrast binary patterns (Fig. 3d, (v)). This compressed image could be directly interpreted by AI models for action recognition, eliminating the need for reconstructing frame sequences. As a result, memory usage and computational overhead were significantly reduced while maintaining recognition accuracy comparable to that of uncompressed video.

Beyond single-mode operation, synaptic vision systems have advanced into unified in-sensor computing architectures that integrate multiple encoding functions within a single device[98]. By tuning synaptic plasticity mechanisms—STP for transient filtering, LTP for associative memory, and controlled decay dynamics for temporal integration—these systems can flexibly switch between contrast enhancement, motion tracking, and pattern recognition (Fig. 3d, (vi)). As a result, the same hardware can highlight object boundaries under low contrast conditions, capture trajectories of moving objects, and recognize complex visual patterns, all of which were achieved by simply reconfiguring its encoding functions. This versatility enabled the extraction of multimodal, task-optimized features at the sensory level, providing AI-optimized data without the need for additional data encoding.

Neuronal vision systems for AI-native robotic vision

Biological neural networks communicate and process information using discrete action potentials, and neurons function as spike-generating units in response to EPSPs that are temporally and spatially integrated at synapses[99–102]. When the EPSP exceeds a threshold, the neuron fires an action potential that is relayed to downstream neurons. Since each action potential consumes only a small amount of energy, this spike-based architecture enables biological vision to operate with exceptional energy efficiency. Moreover, the digitized nature of action potentials ensures robust signal transmission that is resistant to noise and signal degradation during propagation.

Neuronal vision systems emulate this firing behavior of neurons, generating spike trains in response to visual stimuli. By encoding information as discrete spikes rather than continuous-valued signals, neuronal vision systems substantially reduce data volume, thereby enhancing data transmission efficiency throughout the vision processing pipeline. Furthermore, spike-based data are inherently compatible with SNNs that offer superior computational efficiency compared to conventional artificial neural networks, including convolutional neural networks (CNNs) and DNNs that process information using floating-point values[103–108]. In this section, we discuss two key in-sensor computing functionalities enabled by neuronal vision systems: spike encoding that converts continuous optical stimuli into discrete spike trains suitable for SNN-based AI computing and event-driven detection that enables dynamic motion interpretation with low latency and power consumption.

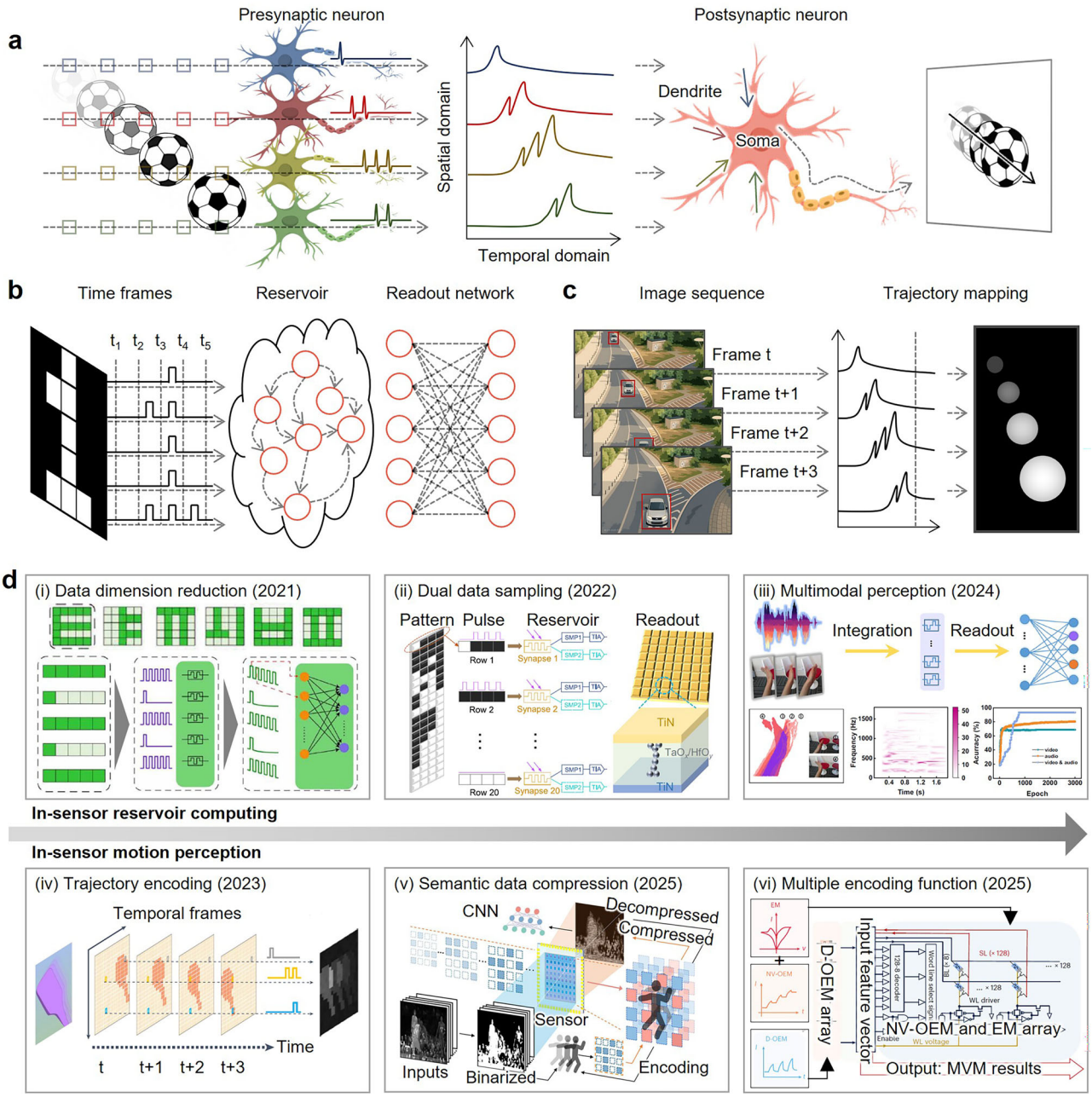


Fig. 3 | Temporal encoding by the synaptic vision system. a Schematic illustration showing temporal summation of EPSPs under repeated presynaptic stimulations, which are then spatially integrated across multiple synapses. **b** Schematic illustration showing reservoir computing performed by synaptic vision systems, which compress high-dimensional visual streams into low-dimensional data. **c** Schematic illustration showing dynamic motion perception using synaptic vision systems, where temporal sequences of frames are encoded into trajectory mapping. **d** Representative demonstrations of reservoir computing and motion perception by synaptic vision systems. The synaptic vision systems reduced data dimensionality by encoding spatial segments into single values (i), with accuracy further improved by incorporating dual sampling strategies (ii) and multimodal perception (iii). These systems also encoded dynamic motions by tracking object trajectories (iv) and compressing semantic frame sequences into binary patterns (v). Finally, multiple encoding functionalities, including contrast enhancement, motion perception, and pattern recognition, were integrated within a single platform (vi). (i) Reproduced under the terms of the Creative Commons CC-BY 4.0 license[93]. Copyright 2021, American Association for the Advancement of Science. (ii) Reproduced under the terms of the Creative Commons CC-BY 4.0 license[94]. Copyright 2022, Springer Nature. (iii) Reproduced under the terms of the Creative Commons CC-BY 4.0 license[95]. Copyright 2024, Springer Nature. (iv) Reproduced with permission[96]. Copyright 2023, Springer Nature. (v) Reproduced under the terms of the Creative Commons CC-BY 4.0 license[97]. Copyright 2025, Springer Nature. (vi) Reproduced with permission[98]. Copyright 2025, Springer Nature.

Spike encoding by neuronal vision systems

Neurons fire action potentials through the spatiotemporal integration of EPSPs generated at synapses. At rest, postsynaptic membrane potential is maintained near −70 mV. When successive EPSPs summate, the membrane potential gradually depolarizes. Once this membrane potential reaches the threshold of ~−55 mV, the neuron fires an action potential; the membrane potential rapidly rises to ~+40 mV, followed by repolarization and hyperpolarization to around −90 mV before returning to its resting state (Fig. 4a). By repeating this cycle, neurons continuously fire action potentials that are transmitted to downstream neurons[99,100].

To emulate this firing behavior, neuronal vision systems integrate spike encoding functionalities into the sensory level, generating spike trains whose

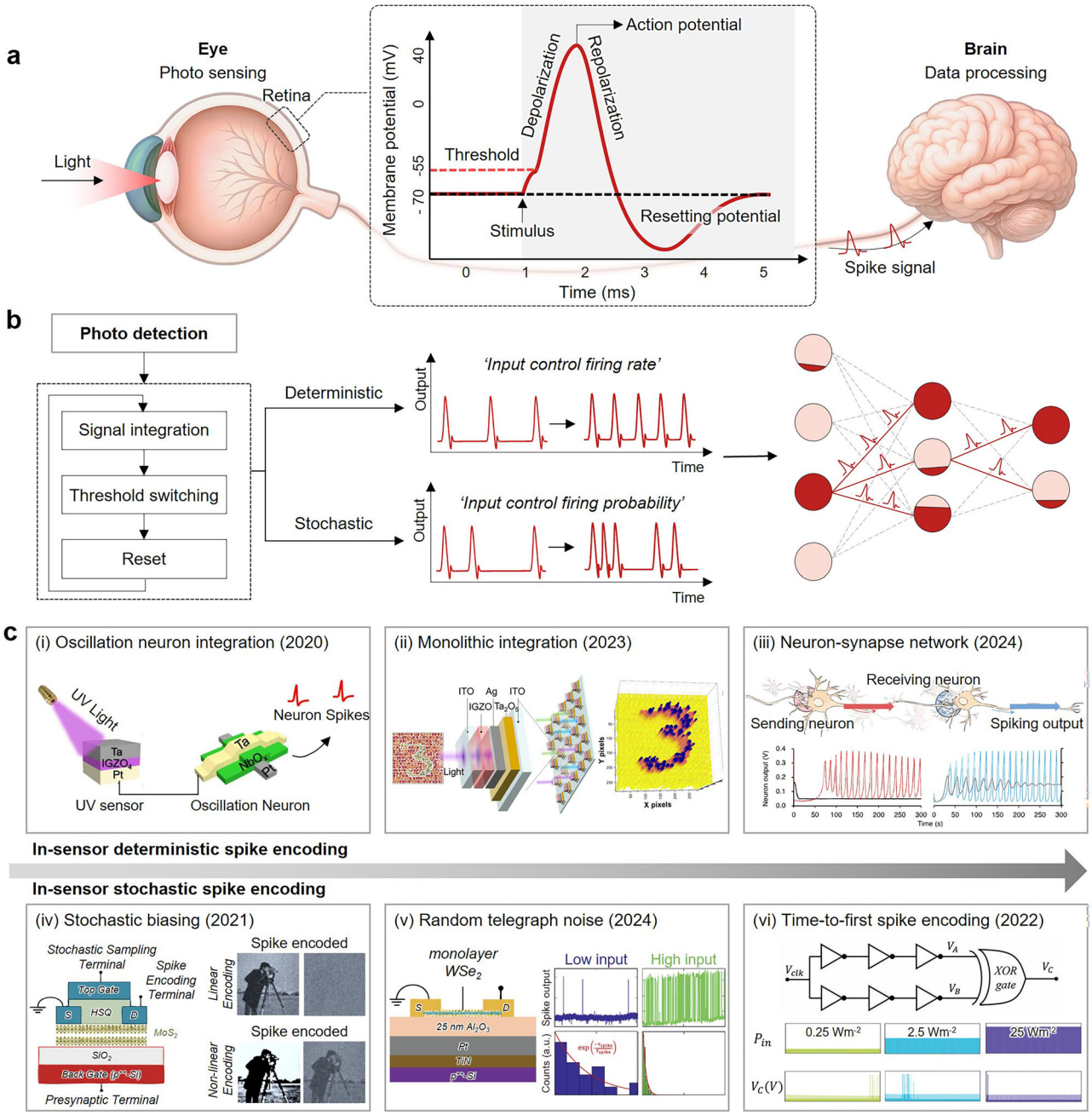


Fig. 4 | Spike encoding by neuronal vision system. a Schematic illustration of action potential firing process in neurons. Neurons communicate with each other using discrete action potentials. **b** Schematic diagram showing in-sensor spike encoding in neuronal vision systems, generating spike trains inherently compatible with SNNs. Spike encoding can be achieved through either deterministic or stochastic methods. **c** Representative demonstrations of deterministic and stochastic spikes encoding by neuronal vision systems. Deterministic spike encoding was achieved by serially connecting nonvolatile memristors with photodetectors (i) or monolithically integrating these components (ii). The spiking devices can be further interconnected using biohybrid synapses, forming a multi-layered neural network (iii). Stochastic spike encoding was achieved by applying random voltage pulses (iv) or exploiting RTN (v). Recently, TTFS encoding was employed to accelerate sensory information processing (vi). (i) Reproduced with permission[111]. Copyright 2020, American Chemical Society. (ii) Reproduced under the terms of the Creative Commons CC-BY 4.0 license[114]. Copyright 2023, Springer Nature. (iii) Reproduced under the terms of the Creative Commons CC-BY 4.0 license[129]. Copyright 2024, Springer Nature. (iv) Reproduced under the terms of the Creative Commons CC-BY 4.0 license[121]. Copyright 2021, Springer Nature. (v) Reproduced under the terms of the Creative Commons CC-BY 4.0 license[122]. Copyright 2024, Springer Nature. (vi) Reproduced with permission[123]. Copyright 2022, Wiley-VCH.

rate, count, or timing is proportional to the intensity of incident light. This in-sensor spike encoding is particularly advantageous for SNN-based vision processing pipelines, which would otherwise require additional software-based data encoding steps to convert floating-point intensity values into spike trains.

Recently, two strategies have been proposed for in-sensor spike encoding: deterministic spiking[109–120] and stochastic spiking[121–125] (Fig. 4b). In deterministic spiking, spiking devices (e.g., NbO_x oscillators and Mott memristors) are integrated with photodetectors to perform signal integration, threshold switching, and reset. Here, electrical signals generated by photodetectors are accumulated until they exceed a threshold, at which point the device fires a spike and subsequently resets to its initial state, generating continuous spike trains with firing rates proportional to light intensity. In contrast, stochastic spiking is achieved by operating the devices

at probabilistic regimes, where the probability of spiking is governed by light intensity. In this regime, the timing of individual spikes is random, whereas the average firing rate and total spike count statistically depend on the light intensity[126–128].

For an example of deterministic spiking, a spiking vision sensor was developed by connecting an ultraviolet (UV)-sensitive photodetector with a NbO_x-based nonvolatile memristor in series (Fig. 4c, (i))[111]. The NbO_x memristor exhibited threshold switching characteristics; it abruptly transitioned from a high-resistance state (HRS) to a low-resistance state (LRS) once the applied voltage exceeded a critical threshold and reset to HRS when the voltage fell below the holding threshold. Therefore, when coupled with a parallel capacitor, this threshold switching enabled periodic charging–discharging cycles that generated oscillatory spike trains. Furthermore, the firing rate depended on the incident light intensity, as stronger illumination increased the photocurrent, shortened the capacitor charging time, and thereby increased the firing rate.

Recently, Wang et al. developed a monolithically integrated spiking vision sensor, in which a threshold-switching memristor was vertically stacked with a photodetector, enabling both light detection and spike encoding within a single device (Fig. 4c, (ii))[114]. In this device, the incident light modulated the conductance of the IGZO photoresistor, which in turn regulated the input current injected into the Ta_2O_3-based memristor. The memristor underwent repetitive switching and subsequently reset due to the charging and discharging of its parasitic capacitance, producing periodic spike trains. Furthermore, the spiking frequency depended not only on intensity but also on wavelength, enabling color-selective photodetection along with in-sensor spike encoding.

Inspired by the cascaded neuron–synapse chains of retinal pathways, Matrone et al. constructed a modular organic circuit capable of adaptively modulating spiking frequency based on both light intensity and prior synaptic activity (Fig. 4c, (iii))[129]. In this system, organic electrochemical transistor (OECT)-based spiking neurons were serially connected with biohybrid synapses. The afferent neurons first converted optical stimuli into spike trains. These spike signals were then transmitted to downstream neurons through biohybrid synapses, with weights electrochemically tuned using neurotransmitters (e.g., dopamine and serotonin). Through redox reactions in the electrolyte, the synapses shifted their baseline conductance, thereby modulating the firing rates of downstream neurons. Therefore, by integrating optical sensing, spike encoding, and synaptic plasticity into a single system, this work demonstrated multilayered neural networks for advanced vision processing.

In contrast to deterministic spiking, stochastic spiking offers an alternative mechanism for spike encoding. A representative example is a spike encoder based on a dual-gate MoS_2 field-effect transistor (FET) connected in series with a Si photodiode. In this configuration, the photodiode generated an analog photovoltage that was applied to the bottom gate of the MoS_2 FET. When the top gate was unbiased, steady but low currents flowed in proportion to illumination. When randomly programmed voltages were applied to the top gate, however, a large current was induced only when the applied bias exceeded a threshold, generating a stochastic spike (Fig. 4c, (iv))[121]. Please note that the firing rate depended on the illumination level. The threshold for spike generation decreased at higher illumination levels, thereby increasing the probability of spike firing. Moreover, by programming the stochastic distribution of input biases, the device could implement multiple encoding capabilities, including spike-rate, spike-count, and spike-timing encoding. Using this encoder, handwritten digit images were encoded as spike trains, which could be directly processed by pretrained SNNs.

Recently, intrinsic material noise, rather than the externally programmed biases, was employed as a stochastic source for spike encoding. For instance, WSe_2 FETs inherently exhibited random telegraph noise (RTN) originating from native point defects such as selenium antisites and tungsten vacancies (Fig. 4c, (v))[122]. These defects induced random fluctuations in channel current, which spontaneously generated stochastic spikes without the need for applying programmed external bias. Importantly, the probability of spike firing increased with the strength of input current, i.e., weak optical and electrical stimuli generated infrequent spikes, whereas stronger inputs yielded a higher spiking rate.

Beyond spike-rate encoding, spike-timing strategies, such as time-to-first-spike (TTFS), were explored to accelerate sensor-level information processing. For TTFS, light intensity is directly translated into the latency of the first spike; brighter illumination generates earlier spikes, whereas weaker illumination results in delayed spikes. Subbulakshmi et al. realized this functionality using an analog photo-encoding circuit, consisting of two cascaded three-stage inverters made of light-sensitive MoS_2 memtransistors, each designed with slightly different switching thresholds, followed by an XOR gate (Fig. 4c, (vi))[123]. Under illumination, the two inverters responded at different times, creating a delay between their outputs. The XOR gate converted this delay into a single spike whose timing depended on the illumination level—shorter delays under strong light generated earlier spikes, while longer delays under weak light yielded later spikes.

Event-driven motion detection by neuronal vision systems

Retinal ganglion cells (RGCs) encode visual information as discrete spike signals. Among them, transient-type RGCs, such as parasol cells in the primate retina, respond selectively to temporal changes in light intensity, including stimulus onset, offset, or motion, while remaining inactive under steady illumination (Fig. 5a)[130–132]. This event-driven firing characteristic filters out redundant static input, reducing the information load and thus enhancing perceptual efficiency, particularly in dynamic motion detection.

Inspired by the transient responsiveness of RGCs to light intensity change, event-driven cameras, often referred to as silicon retina such as dynamic vision sensor (DVS) and dynamic and active-pixel vision sensor (DAVIS), have been developed[133–138]. Unlike conventional image sensors that synchronously capture and transfer full image frames at fixed intervals, event-driven cameras operate asynchronously at the pixel level, generating spike signals only when local changes in light intensity exceed a threshold (Fig. 5b). Therefore, these cameras can selectively capture motion-related features while bypassing redundant static information, reducing data transmission volume to processors and thus enabling low-latency, energy-efficient detection of dynamic motion[139–141].

However, conventional event-driven cameras typically rely on complex pixel architectures incorporating multiple analog components, such as amplifiers, differentiators, and comparators, to detect temporal changes in light intensity and generate spike outputs when the threshold is exceeded. These pixel architectures increase the pixel footprint. As a result, recent efforts have focused on developing event-driven vision systems based on new materials and device structures that inherently generate event-driven outputs at the sensory level. These devices can realize DVS- or DAVIS-like operation within simplified architectures, minimizing circuit overhead while providing motion-related features that AI models can efficiently process[142–144].

For instance, an event-driven pixel was developed by connecting two photodiodes in opposite directions, with a RC circuit integrated into one of the branches to induce asymmetric temporal photoresponse—specifically, differing rise and decay times (Fig. 5c, (i))[145]. When light intensity changed, the mismatch in photoresponse speed between fast and slow photodiodes generated a transient differential current, corresponding to a spike output. Under steady illumination, the opposing photocurrents canceled each other out, yielding zero current. This device-level approach enabled event-driven spike generation without the need for complex analog circuits, enabling energy-efficient motion detection by AI computing.

Additionally, a photosensitive metal–insulator–semiconductor (MIS) structure was proposed to enable event-driven operation within a single device[146–150]. In this device, the insulating layer was inserted to form a capacitive architecture, which suppressed steady-state current by reaching equilibrium under constant illumination while allowing transient charging and discharging currents to flow only when light intensity changed (Fig. 5c, (ii))[148]. As a result, spike signals were exclusively generated during light-on and light-off transitions, enabling bidirectional event encoding without the

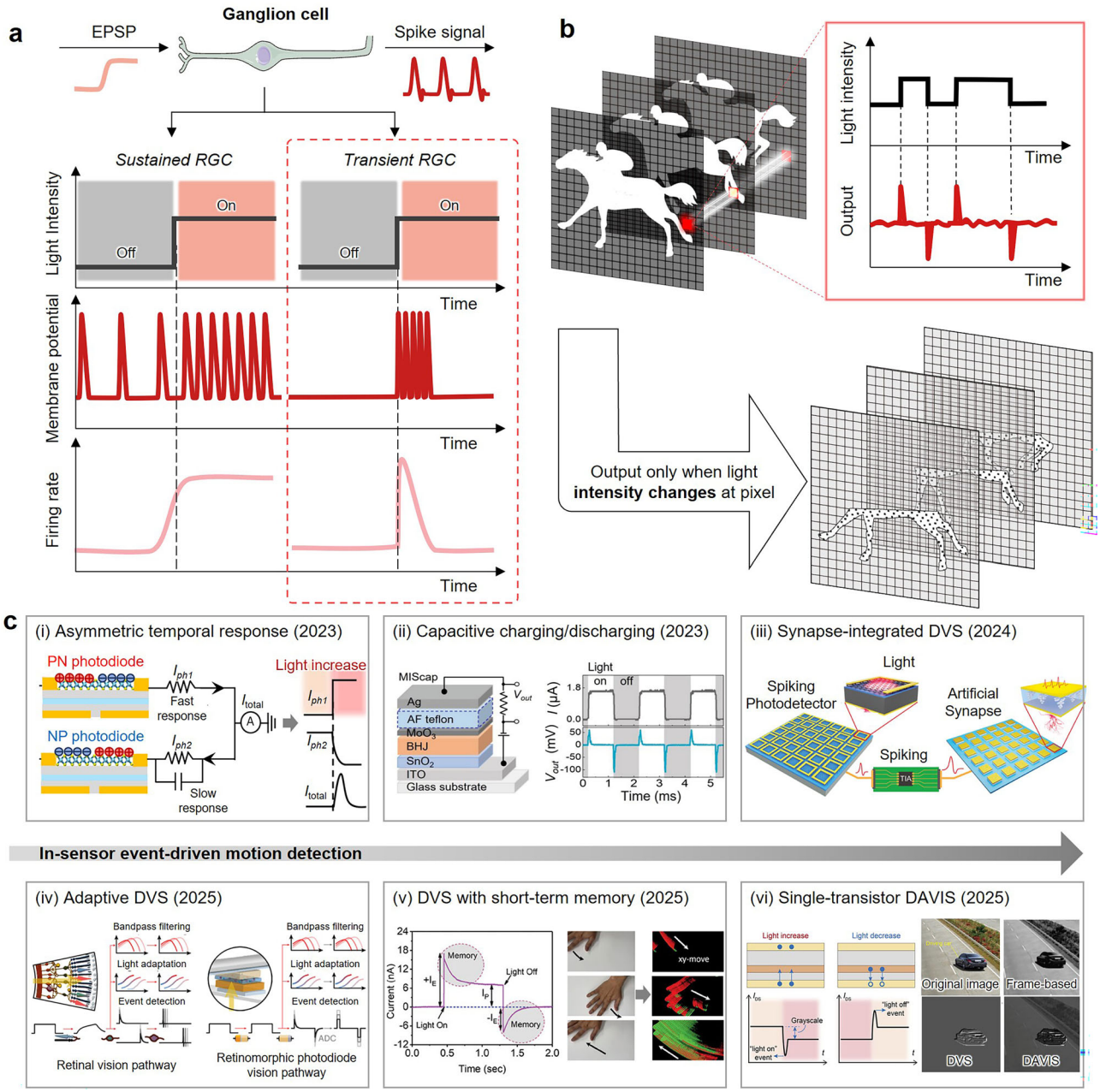


Fig. 5 | Event-driven motion detection by a neuronal vision system. a Schematic illustration of event-driven spiking in transient RGCs that respond selectively to changes in light intensity. **b** Schematic illustration of event-driven motion detection by neuronal vision systems, where spike signals are exclusively generated at pixels in which light intensity changes. **c** Representative demonstrations of event-driven motion detection by neuronal vision systems. Event-driven operation was achieved by connecting two photodetectors with asymmetric temporal responses (i) and introducing metal–insulator–semiconductor capacitive structures (ii). The integration of additional functionalities, such as synaptic connection (iii), environmental adaptivity (iv), short-term memory (v), and DAVIS capability (vi), provided richer spatiotemporal information for accurate AI inference. (i) Reproduced with permission[145]. Copyright 2023, Springer Nature. (ii) Reproduced under the terms of the Creative Commons CC-BY 3.0 license[148]. Copyright 2023, Wiley-VCH. (iii) Reproduced with permission[149]. Copyright 2024, Wiley-VCH. (iv) Reproduced with permission[150]. Copyright 2025, Springer Nature. (v) Reproduced with permission[152]. Copyright 2025, Wiley-VCH. (vi) Reproduced under the terms of the Creative Commons CC-BY 3.0 license[153]. Copyright 2025, Wiley-VCH.

need for multiple device components per pixel and thus offering a potential for high-density event-driven vision systems[151].

Recently, MIS-based event-driven sensors were extended to emulate not only retinal signal encoding, but also synaptic integration observed in cortical processing. Wu et al. proposed a vision system that replicated the human visual pathway through the physical connection of an event-driven image sensor and an artificial synapse (Fig. 5c, (iii))[149]. The spike current generated by the image sensor was converted to voltage via a transimpedance amplifier (TIA) and delivered to the artificial synapse, where it was potentiated or depressed depending on spike polarity and timing. This architecture enabled real-time training of the artificial synapses, allowing the system to achieve recognition of spatial and temporal patterns, including symbolic images, hand gestures, and motion sequences.

Although those event-driven vision systems offered energy-efficient motion detection methodology by selectively capturing dynamic features, they often fell short in providing the visual details required for high-level visual recognition tasks. Because they only encoded temporal contrast, event-driven outputs often missed fine structural cues, such as texture and

shape, which were essential for accurate object recognition. To address these limitations, recent works have focused on integrating additional sensing functionalities that enable more context-aware data acquisition, thereby improving compatibility with AI-based inference.

One example is the integration of adaptive functionality into an event-driven vision system. Lin et al. developed a donor–acceptor heterojunction photodiode with a Schottky interface and an ion reservoir (Fig. 5c, (iv))[150]. This device generated spike trains in response to light intensity changes due to the Schottky barrier. Under the sustained illumination, however, $TFSI^-$ anion in the ion reservoir drifted into the p-type semiconductor layer and increased the junction barrier height, leading to a decrease in spike amplitude. Therefore, signal saturation can be prevented by dynamically modulating the amplitude level based on stimulus duration, thereby realizing the environment-adaptive event-driven vision system capable of reliable motion detection under high dynamic range conditions.

In addition, short-term memory characteristics were integrated into event-driven vision systems to enhance spatiotemporal perception (Fig. 5c, (v))[152]. By symmetrically inserting a high-*k* dielectric layer between the photodiode channel and electrodes, light-induced charge accumulation and its gradual release were achieved, enabling a transient memory effect in the spike outputs. In contrast to the aforementioned synaptic photodetectors that captured afterimages from both dynamic and static regions (Fig. 3c, (i)), this device selectively extracted the temporal trace of a moving object while suppressing the static background. As a result, only motion-induced stimuli were preserved as trajectory-relevant information, enabling the perception of object movement without interference from stationary scenes.

Recently, Feng et al. developed a charge-coupled phototransistor capable of detecting both dynamic and static visual information, similar to the functionality of DAVIS. While DAVIS typically requires 15–50 transistors per pixel[133,134,153–155], this device achieved the same functionality with a single-transistor configuration. This device consisted of two photosensitive capacitors with asymmetric dielectrics, one enabling rapid charge tunneling during illumination onset and offset, and the other exhibiting charge-blocking behavior under sustained exposure. This asymmetric charge dynamics allowed the device to generate event-driven spikes in response to light intensity changes, followed by a steady output proportional to the incident light intensity (Fig. 5c, (vi))[156]. Consequently, this device simultaneously detected both transient events and static scene details, offering a compact alternative to the conventional DAVIS designs.

Hierarchical vision systems for AI-native robotic vision

In the human retina, various types of retinal cells are hierarchically interconnected; multiple photoreceptors (e.g., rods and cones) converge onto bipolar cells, which in turn converge onto RGCs. Visual signals are progressively integrated through these hierarchical networks, enabling vision processing at the retinal level before signals are transmitted to the visual cortex[157,158]. One key functionality of this retinal hierarchy is the center-surround receptive field, which enhances spatial contrast via antagonistic responses to light stimuli[159]. Upon illumination, photoreceptors transduce photon energy into the graded membrane potential of bipolar cells[160]. Bipolar cells are classified into ON and OFF types that respond selectively to light and darkness, respectively. These ON and OFF bipolar cells converge onto the same RGC in an antagonistic center-surround configuration, allowing the RGC to integrate excitatory and inhibitory inputs within its receptive field (Fig. 6a). As a result, RGCs fire action potentials with heightened sensitivity to spatial features, such as object edges and local luminance variations.

Inspired by the hierarchical networks—particularly the center-surround receptive fields—hierarchical vision systems have been developed to perform spatial image processing at the sensory level[48]. These systems employ device-level kernels, in which positive and negative weights are physically implemented, enabling excitatory and inhibitory integration of visual stimuli. As a result, feature-oriented operations, such as filtering and convolutional operations, can be executed through in-sensor computing. For example, hierarchical vision systems with edge detection kernels can directly capture edge-extracted images, reducing the data volume to be transferred and processed by AI models (Fig. 6b). Furthermore, by programming kernel weights to match those used in CNNs, convolutional operation can be performed at the sensory level, generating feature maps optimized for downstream AI computing (Fig. 6c).

Wang et al. developed a hierarchical vision system by arranging multiple phototransistors with opposite photoresponse polarities—an OFF-type device at the center surrounded by ON-type devices at the periphery (Fig. 6d, (i))[161]. This configuration emulated the excitatory and inhibitory configuration of biological center-surround receptive fields, forming a device-level kernel that performed analog integration of central and peripheral signals to directly extract spatial contrast. Moreover, by modulating the gate bias of each device, the kernel weights could be reconfigured to perform diverse filtering operations, including edge detection, image stylization, and contrast correction, through in-sensor computing.

Spectral selectivity was also incorporated into hierarchical vision systems to enable wavelength-selective filtering[162–164]. Pi et al. demonstrated a broadband kernel device using $PdSe_2/MoTe_2$ van der Waals heterojunction, where the constituent materials exhibited complementary absorption spectra, from UV to near-infrared (Fig. 6d, (ii))[162]. By modulating its band alignment via gate bias control, the relative photoresponse of each spectral band could be tuned at the pixel level, enabling spectral decomposition of incoming images. These wavelength-specific components could then be filtered by the kernel device to emphasize spectral features, such as edges or sharpness within the target bands.

Scene-adaptive functionality was also introduced to improve filtering capabilities, aiming to detect low-visibility targets under low-light and low-contrast conditions[165]. It was achieved using dynamic kernels—rather than static kernels—in which the photoresponsivity of a central active pixel was modulated according to the electrical outputs generated from surrounding passive pixels via circuitry-based voltage adjustment (Fig. 6d, (iii)). This configuration amplified subtle luminance differences that were often imperceptible in conventional static kernels with fixed weights, while suppressing uniform, non-informative backgrounds. Consequently, this dynamic kernel adaptively reconfigured its weights according to the input scene, achieving high-contrast edge extraction for robust target tracking even under challenging illumination.

In another application of hierarchical vision systems, an array of responsivity-tunable photodiodes was used to execute a multiplication-accumulation (MAC) operation directly in the optoelectronic domain[166]. In this device, the responsivity of each photodiode was individually adjusted via gate bias control to encode both positive and negative weights. Upon the irradiation of light patterns, each pixel generated a photocurrent proportional to the product of the incident light intensity and its programmed responsivity. The pixel currents were then summed to yield the weighted sum of input intensities that should be otherwise computed through digital MAC operations (Fig. 6d, (iv)). By executing MAC operations in the analog domain, pattern classification could be achieved at the sensory level, while avoiding data transfer and processing bottlenecks of conventional AI computing procedures.

To enable persistent kernel configurations, non-volatile memory capability was integrated into hierarchical vision systems[167–169]. Cui et al. developed a photosensor array based on a ferroelectric $Pb(Zr_{0.2}Ti_{0.8})O_3$ layer, in which the remanent polarization state determined the pixel responsivity (Fig. 6d, (v))[167]. In this device, the responsivity could be programmed by controlling the polarization state; positive responsivity for up-polarization and negative responsivity for down-polarization. Once programmed, the responsivity remained stable for a long period (>1000 s), despite the absence of external biasing, ensuring non-volatile kernel configurations. Therefore, the need for repeated kernel reprogramming was eliminated, reducing power consumption and latency for in-sensor MAC operations.

Beyond traditional planar architecture, vertically integrated kernels were developed to execute multiple types of MAC operations in parallel (Fig. 6d, (vi))[170]. Each stacked layer functioned as an independent kernel with

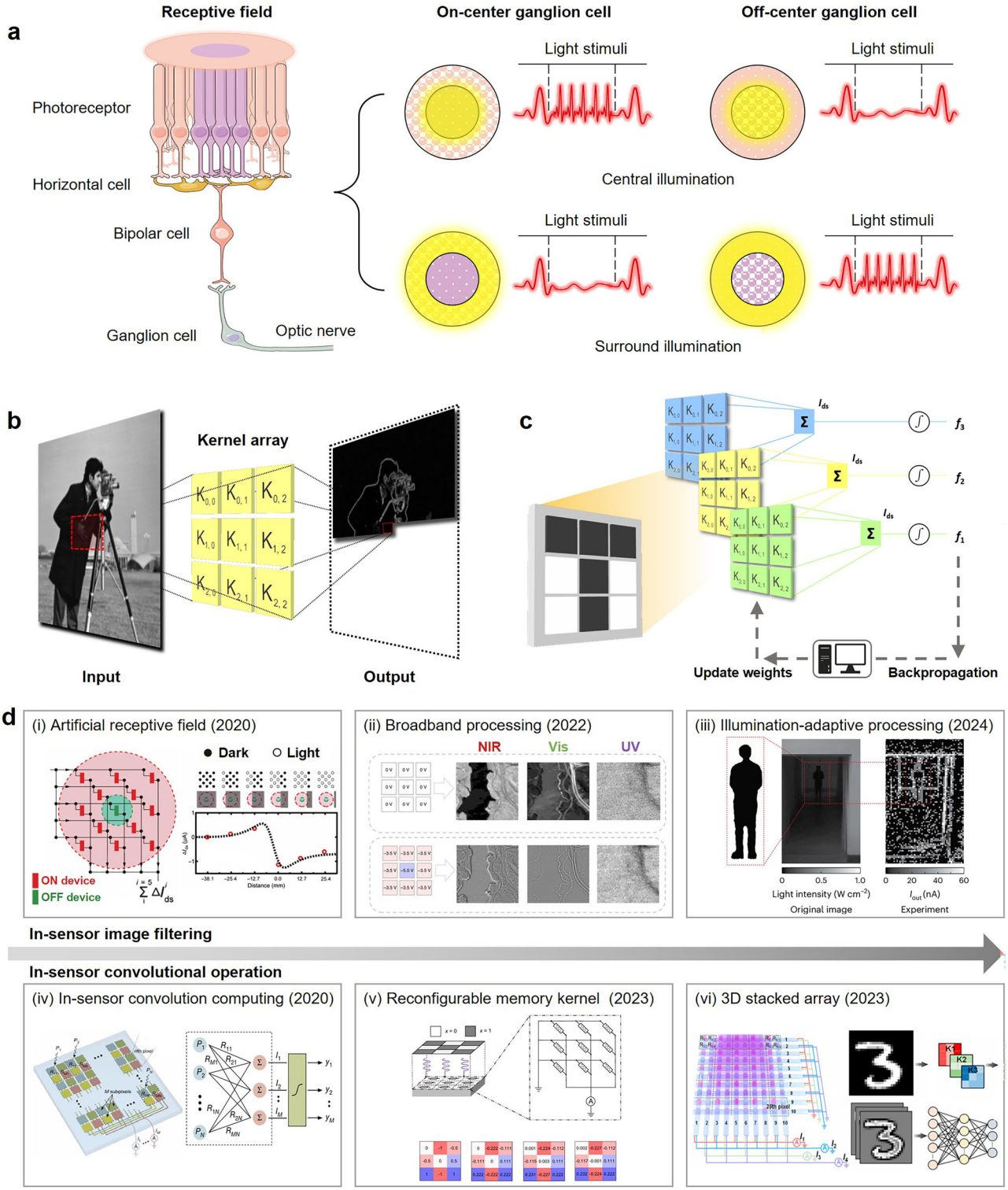


Fig. 6 | Image filtering and convolutional operation by a hierarchical vision system. a Schematic illustration of center-surround receptive fields in biological retina, where the spatial distribution of illumination modulates action potential firing in on-center and off-center RGCs. **b** Schematic illustration showing image filtering performed by hierarchical vision systems, achieved through MAC operations between input images and filtering kernels. **c** Schematic illustration of in-sensor convolutional operation, where MAC operations for CNNs are executed at the sensory level. **d** Representative demonstrations of image filtering and convolutional operation by hierarchical vision systems. Hierarchical vision systems formed artificial receptive fields within devices (i), with capabilities further extended to broadband processing (ii) and illumination-adaptive processing (iii). Furthermore, MAC operations between trained CNN kernels and incident light were achieved through in-sensor computing (iv), further improved by incorporating memory effect for low-power consumption (v) and 3D stacked configurations for parallel MAC operations (vi). (i) Reproduced under the terms of the Creative Commons CC-BY license[161]. Copyright 2020, American Association for the Advancement of Science. (ii) Reproduced with permission[162]. Copyright 2022, Springer Nature. (iii) Reproduced with permission[165]. Copyright 2024, Springer Nature. (iv) Reproduced with permission[166]. Copyright 2020, Springer Nature. (v) Reproduced with permission[167]. Copyright 2023, Springer Nature. (vi) Reproduced with permission[170]. Copyright 2023, American Chemical Society.

uniquely programmed weights, allowing the simultaneous extraction of diverse feature maps, such as edge features in vertical, horizontal, and diagonal orientations. As a result, the feature maps necessary for AI computing could be acquired through a single-readout operation, eliminating repetitive in-sensor MAC operations and thus improving processing speed.

Challenges for AI-native robotic vision

Despite recent progress, practical employment of in-sensor computing remains limited by several key challenges. In this section, we discuss the major factors that hinder the translation of current laboratory-scale demonstrations into a practical robotic vision system.

Scalability

Scalability represents one of the most critical bottlenecks in the commercialization of robotic vision systems based on in-sensor computing. Most existing works have been restricted to single devices or small-scale arrays, whereas real-world robotic vision typically demands wafer-scale integration of millions of pixels operating with consistent performance and reliability. To meet such demands, breakthroughs in pixel miniaturization, fabrication yield, and active-matrix addressing are required, together with the establishment of CMOS-compatible, wafer-level manufacturing processes suitable for high-throughput production. Furthermore, the manufacturing processes should ensure electrical and optical uniformity across large-scale arrays for reliable and reproducible system-level operation.

Bandwidth

The temporal dynamics of in-sensor computing should be improved to meet the bandwidth requirements of real-time robotic applications. In particular, the in-sensor computing cycle for each frame should be shorter than or comparable to the frame interval of a conventional robotic vision system to prevent temporal information loss and preserve the fidelity of dynamic scene perception. However, current in-sensor computing techniques remain far from this regime. For instance, synaptic vision systems typically require long signal accumulation periods, ranging from several hundred milliseconds to a few seconds, to generate a single frame. Neuronal vision systems often fail to encode multiple spikes within each frame interval, and hierarchical vision systems face challenges in executing convolutional operations and patch-wise processing at high speed. To achieve frame rates comparable to commercial ones, therefore, both material-level innovations and circuit-level co-design strategies are necessary to support high-bandwidth operation without compromising synaptic, neuronal, and hierarchical functionalities.

Applications

While most in-sensor computing studies have focused on device-level development, the ultimate goal is their system-level integration into practical platforms (e.g., autonomous vehicles and mobile robots), enabling low-latency and energy-efficient perception in real-world environments[139]. However, existing works often showed proof-of-concept demonstrations, such as handwritten digit classification or basic motion detection, which fail to reflect the complexity of dynamic and unpredictable robotic environments. Therefore, in-sensor computing technologies should advance toward real-world applicability and, at least, validate their feasibility for system-level integration. Moreover, a standardized evaluation framework is necessary for fair cross-device comparison. Such benchmarking protocols should systematically evaluate latency, energy efficiency, dynamic range, endurance, and task-level accuracy under realistic and unstructured conditions.

Co-design with AI models

Unlike conventional image sensors that capture discrete image frames with pixel intensity values, synaptic, neuronal, and hierarchical vision systems generate fundamentally different data formats, such as temporally accumulated signals, asynchronous spike events, and local feature maps. As a result, it is often ineffective to apply such unconventional data to conventional AI models, necessitating the co-design of AI architectures optimized for each modality. For instance, event-driven vision systems necessitate SNN models capable of processing asynchronous temporal information. These hardware-algorithm co-designs will enable energy-efficient and real-time perception frameworks, paving the way toward the practical realization of AI-native robotic vision.

Conclusion

In-sensor computing has emerged as an essential technique for AI-native robotic vision, enabling the direct generation of AI-optimized visual data that supports efficient and accurate inference. In particular, recent advances in synaptic, neuronal, and hierarchical vision systems have accelerated this technological progress, reducing the burdens of conventional image signal processing and thus enhancing the efficiency of data transmission and processing. For instance, synaptic vision systems extract spatial and temporal features by selectively amplifying semantically relevant signals and compressing temporally correlated patterns. Neuronal vision systems generate spike-based outputs inherently compatible with SNNs and also support event-driven motion detection. Hierarchical vision systems execute image filtering and convolutional operations using sensor-level kernels, thereby streamlining AI inference pipelines.

The future of robotic vision systems should integrate synaptic, neuronal, and hierarchical functionalities within a single device architecture, where multiple encoding modalities coexist and can be selectively activated according to task demands. In such integrated systems, synaptic preprocessing would suppress noise and enhance contrast, neuronal modules would encode sensory inputs into sparse spike trains, and hierarchical kernels would extract spatial features, all of which would be executed at the sensory frontend upon the AI-driven requirements. This integration promises robotic vision systems with dynamic reconfigurability, enabling adaptive perception optimized for diverse tasks. Ultimately, this in-sensor computing will offer the foundation for future robotic platforms with human-like perception and adaptability, enabling efficient and robust robotic operations in real-world environments.

Data availability

No datasets were generated or analyzed during the current study.

Received: 1 October 2025; Accepted: 25 November 2025;
Published online: 07 January 2026

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Acknowledgements

This research was supported by IBS-R006-A1. This research was also supported by the Korea Evaluation Institute of Industrial Technology (KEIT) grant funded by the Korean government (MOTIE) (00420534), by the Ministry of Culture, Sports and Tourism (MCST) of Korea and the Korea Creative Content Agency (KOCCA) as part of the Culture Technology (CT) Research & Development Program (RS-2024-00439361), and the Future Resource Research Program (2E33542) of the Korea Institute of Science and Technology (KIST).

Author contributions

C.C. and D.-H.K. supervised this project. D.K., J.I.K., and Y.K. prepared the figures. D.K., J.I.K., Y.K., C.C., and D.-H.K. wrote the manuscript. All authors reviewed and approved the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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